

Fair-EAS: Entanglement Allocation and Selection for Process-Oriented Fairness in Quantum Communication Networks

Zhaoying Wang¹, Student Member, IEEE, Jian Li², Senior Member, IEEE,
Kaiping Xue³, Senior Member, IEEE, Zhonghui Li⁴, Member, IEEE,
Ruidong Li⁵, Senior Member, IEEE, Nenghai Yu⁶,
Qibin Sun⁷, Fellow, IEEE, and Jun Lu⁸

Abstract—Quantum communication networks enable advanced quantum applications through remote entanglement distribution among source-destination pairs. Despite efforts to optimize entanglement distribution, fairness in multi-request scenarios has been neglected, potentially causing issues like “request starvation”. To address such issue, this paper concentrates on the unique properties of entangled systems and introduces a process-oriented fairness metric, i.e., expected throughput, departing from conventional approaches used in classical networks. Furthermore, we propose an entanglement distribution scheme named Fair-EAS, which prioritizes entanglement allocation and selection for batching multiple requests to maximize overall throughput while maintaining max-min fairness. To facilitate a convenient solution, we transform the nonlinearity of the problem into an equivalent linear programming formulation and decouple the solution into offline and online phases. In the offline phase, we design a multi-round water-filling-like optimization algorithm to determine the optimal path set for predicting entanglement allocation. In the online phase, we introduce an adaptive compensation algorithm and an entanglement “fragment” exhaustion algorithm to dynamically adjust the path set based on successfully generated entangled pairs. Comprehensive simulations show that Fair-EAS outperforms the existing schemes in terms of fairness by significantly enhancing the minimum throughput and throughput deviation among multiple requests while maintaining an overall throughput close to the optimal level.

Index Terms—Max-min fairness, entanglement distribution, quantum communication networks.

I. INTRODUCTION

IN RECENT years, quantum communication networks (QCNs) have gained significant attention due to their foundation in quantum physics and unique ability to generate, process, and locally store quantum information within quantum nodes [1]. As a brand new infrastructure, QCNs can achieve end-to-end (E2E) entanglement distribution across the large-scale network, and further provide the physical implementation of advanced quantum applications, such as secure quantum key distribution [2], clock synchronization [3], [4], and distributed quantum computing [5], [6].

The functional foundation of QCNs is to share entangled pairs between any pair of stations. To obtain E2E entangled pairs efficiently, related studies have been dedicated to design entanglement distribution schemes in various scenarios to improve the distribution performance, such as improving the throughput [7], [8], [9], centralized [10] or distributed routing protocols [11], resource allocation [12], [13], request scheduling [14], entanglement quality guarantee [15] and so on. However, as an important issue, researchers rarely consider fairness in the existing studies. However, in the network field, fairness is an indispensable aspect that researchers need to focus on when improving the performance on other metrics. The lack of consideration of fairness may lead to two lousy results in practical implementation, especially for dealing with multiple requests in large-scale QCNs. Firstly, at the network level, poor fairness leads to load imbalance. For example, too many bottleneck links cause network performance degradation, such as reduced throughput and increased distribution latency. Secondly, at the user level, unfairness inevitably leads to “starvation”, which means some non-dominant requests are not served for a long time or not fully satisfied. Most quantum applications, such as distributed quantum computing [16], quantum clock synchronization [17], have a necessary requirement on a certain quantity of E2E available entangled pairs. “Starvation” causes these applications to lose their normal functionality. Therefore, it is necessary to consider carefully fairness in QCNs.

Received 19 February 2024; revised 1 September 2024 and 3 November 2024; accepted 1 December 2024. Date of publication 12 December 2024; date of current version 17 July 2025. This work is supported in part by the Innovation Program for Quantum Science and Technology under Grant No. 2021ZD0301301, Anhui Initiative in Quantum Information Technologies under grant No. AHY150400, National Natural Science Foundation of China under Grants No. 62402466, Youth Innovation Promotion Association of Chinese Academy of Sciences (CAS) under Grant No. Y202093, and Japan Society for the Promotion of Science (JSPS) KAKENHI under Grant No. 23K28070. The associate editor coordinating the review of this article and approving it for publication was M. Levorato. (Corresponding authors: Jian Li; Kaiping Xue.)

Zhaoying Wang, Jian Li, Kaiping Xue, Zhonghui Li, Nenghai Yu, and Qibin Sun are with the School of Cyber Science and Technology, University of Science and Technology of China, Hefei, Anhui 230027, China (e-mail: lijian9@ustc.edu.cn; kpxue@ustc.edu.cn).

Ruidong Li is with the Institute of Science and Engineering, Kanazawa University, Kakuma, Kanazawa 920-1192, Japan.

Jun Lu is with the School of Cyber Science and Technology and the Department of Electronic Engineering and Information Science, University of Science and Technology of China, Hefei, Anhui 230027, China.

Digital Object Identifier 10.1109/TCOMM.2024.3516497

Unfortunately, the unique features of quantum mechanics, e.g., quantum decoherence [18] and probabilistic properties, introduce significant challenges in designing fairness-guarantee entanglement distribution schemes for QCNs. The first challenge arises from the incompatibility of the standard fairness analysis model used in classical networks. E2E entanglement distribution involves generation on quantum links between adjacent nodes and swapping on repeaters along the reserved path [19], [20], [21]. The inherent probabilities associated with these processes render traditional fair bandwidth allocation models meaningless in QCNs [22]. In classical networks, nodes typically expect fair bandwidth allocation, coupled with quality of service (QoS) considerations [23]. An example is that a path with more hops is more challenging to distribute E2E entangled pairs due to more swapping operations. Therefore, classical fairness-solving algorithms are ill-suited for QCNs, demanding the formulation of different fairness models and quantitative fairness metrics.

Another challenge lies in the timeliness of algorithmic solutions, essential to counteract decoherence—a process where coherence is lost over time due to interactions between the quantum system and the external environment [24]. For instance, in the most advanced experimental studies to date, effective storage times range from nanoseconds to milliseconds [25], depending on various entanglement preparation techniques and storage media such as the DLCZ scheme [26] and nitrogen-vacancy (NV) centers [27]. Therefore, due to the developmental limitations of current physical device technology, the effective lifetime of entangled pairs remains very short. Accordingly, effective solutions should avoid complex and time-consuming decisions for routing and resource allocation after entanglement generation. Otherwise, entangled pairs inevitably lose efficacy without preserving quantum characteristics. Addressing these aforementioned challenges is essential for achieving fair entanglement distribution, necessitating innovative approaches beyond classical algorithms.

In this paper, we address the above challenges associated with entanglement distribution by undertaking a comprehensive remodeling and quantification of metrics for process-oriented fairness. Our focus is on quantifying the expected throughput after entanglement distribution, with the ultimate goal of achieving max-min fair results among multiple requests. Given the unique entanglement properties and the inherent complexity of the solution, we design both offline and online phases. In the offline phase, a prediction algorithm is strategically deployed before entanglement generation, aiming to predict allocation results on nodes and links for each request. This phase is crucial for laying the groundwork for a fair allocation of entanglement resources. In the online phase, a resource allocation algorithm, characterized by low time overhead, proves highly operational in compensating for any deviations in resource allocation arising from actual resource generation. Moreover, the online algorithm is adept at addressing the challenge of the exhaustion of “fragments”, which refers to entangled pairs existing in the residual network but not occupied by any requests. By adopting a systematic approach that considers both offline and online aspects, our

design and consideration of various factors contribute to the effectiveness and reliability of our proposed solution.

The main contributions of this paper are summarized as follows:

- 1) As the pioneering exploration of process-oriented fairness in entanglement distribution, we quantify expected throughput as a novel fairness metric. We model the max-min fairness problem among multiple requests' expected throughput and transform the inherent nonlinearity of the problem into an equivalent integer linear programming formulation.
- 2) We propose Fair-EAS, an offline-online stepwise solution for better compatibility. In the offline phase, we employ the multi-round water-filling-like optimization (MWF) algorithm to predict the optimal path set for each request. We present an adaptive compensation (ACE) algorithm of successfully generated entangled pairs and an entanglement “fragment” exhaustion (EFE) algorithm online to enhance both fairness and throughput.
- 3) Through comprehensive simulations, we assess the performance of Fair-EAS and existing schemes. Our observations demonstrate that Fair-EAS ensures the optimal fairness, in terms of the minimum throughput, throughput deviation, and Jain's index, while maintaining a close-optimal overall throughput.

The rest of this paper is organized as follows. We list and discuss several relevant researches in Section II. We introduce the preliminary knowledge and research motivation in Section III. The network model, the quantification and definition of fairness are described in Section IV. Each algorithm of our Fair-EAS scheme and analysis is described in detail in Section V. After that, in Section VI, we finish the simulation to verify and analyze the effectiveness of Fair-EAS scheme compared with other existing schemes. Finally, we draw the conclusion and present potential future work in Section VII.

II. RELATED WORK

Amidst the revolutionary development of QCNs, the urgent demand for efficient and scalable entanglement distribution schemes has intensified [28], [29], [30]. This section provides an overview of related work on routing and resource allocation in QCNs.

Entanglement routing fundamentally defines path selection for any source-destination (SD) pair using various routing metrics. Initially, Van Meter et al. [31] explored the application of the Dijkstra algorithm, defining optimal link costs to find a single path for each SD pair in QCNs, which is a fundamental routing scheme. Pant et al. [32] introduced a greedy algorithm in a grid QCN, utilizing paths with the fewest hops to establish entangled pairs based on information about all successfully generated entangled pairs. While this multipath routing scheme improves performance compared to a single-path scheme, the decisions based only on the fewest hops cannot guarantee the optimal distribution performance.

Shi and Qian [33] proposed Q-CAST scheme, utilizing the extended Dijkstra's algorithm to find the most reliable paths (i.e., maximum expected throughput) by incorporating the success probability of entanglement generation and swapping.

Q-CAST seeks auxiliary sub-paths to enhance the performance of the main path. However, Q-CAST compensates too late for failures, getting blocked with established sub-paths, and lacks flexibility in utilizing successfully created entangled pairs to connect these sub-paths. Similar to routing related to quality of service, to evaluate the fidelity (i.e. quality) of entanglement, Li et al. [15] proposed purification-enabled entanglement routing designs to provide fidelity guarantee for multiple SD pairs in QCNs. This work designs an extended Dijkstra algorithm, Q-LEAP, to reduce the computational complexity by using a simple but effective purification decision.

Besides path selection, entanglement routing invariably involves resource allocation issues, focusing on contention-free and fair allocation for multiple requests on shared links. Li et al. [13] proposed three heuristic algorithms (i.e., PS: Proportional Share, PF: Progressive Filling, PU: Propagatory Update) for scheduling capacity allocation on the public edges in QCNs. While easily deployable in distributed network settings, these algorithms address fair allocation at the link-layer level and do not cater to the more practical fair E2E entanglement distribution.

Furthermore, Zhao and Qiao [10] introduced REPS routing algorithm, aiming to provide necessary network resources, subject to constraints, for both maximum throughput and built-in redundancy. However, this work primarily focuses on improving total throughput, without discussion on whether individual requests are satisfied in concurrent multi-request scenarios. The effectiveness of entanglement distribution for non-dominant requests, especially marginal users, in large-scale scenarios remains uncertain.

In summary, the majority of existing solutions overlook the aspect of process-oriented fairness in entanglement distribution within QCNs, potentially leading to pronounced instances of request “starvation” when managing multiple requests. Simultaneously, the presence of heavily congested links, i.e., a quantum node or link is carrying less entanglement than the amount of requests for it to handle, contributes to the degradation of distribution performance for affected requests. Subsequently, we visually illustrate the distinctions among these solutions in a representative topology, highlighting their corresponding performances.

III. BACKGROUND AND MOTIVATION

This section introduces the key entanglement technical features and the necessary network settings for easy follow-up. After that, we explain the motivation behind our study using a typical example.

A. Preliminary Knowledge

Generally, the distribution of E2E entanglement primarily encompasses two processes. Entanglement generation is used to prepare and distribute short-distance entangled pairs between adjacent nodes. An entanglement generation source is constructed in the middle of the link or at the node. Then the entangled qubits are respectively sent to adjacent nodes through quantum channels and stored into their respective quantum memory. Due to very inefficient long-distance direct

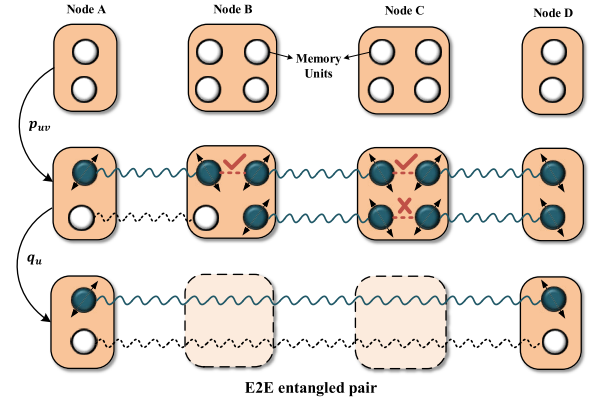


Fig. 1. Key operations of distributing remote entangled pairs: entanglement generation and entanglement swapping.

entanglement generation, we use entanglement swapping to extend short-distance point-to-point entanglement to long-distance end-to-end entanglement. Specifically, we prepare two short-distance entangled pairs and then perform Bell-state measurement on one particle of the first pair and one particle of the second pair. Based on the measurement results, apply corresponding quantum operations to the remaining two particles, which enables these two originally unentangled particles to establish a new entanglement relationship, thereby completing entanglement swapping and achieving long-distance entanglement [34], [35]. However, the inherent imperfections of quantum devices introduce uncertainty into quantum gate/measurement operations [36], [37]. Consequently, the generation and swapping of entanglement are inherently probabilistic processes, and their respective success probabilities can be denoted as p_{uv} on edge (u, v) and q_u on node u (both less than 1).

Fig. 1 illustrates an end-to-end path in QCNs consisting of four nodes, labeled A-B-C-D. At each node, dedicated memory units are reserved for storing the subsequently successfully generated entangled pairs. The number of memory units indicates the width of the path. The width of this path is 2 (where the end nodes are allocated 2 memory units, and the repeaters are allocated 4 memory units due to the bidirectional entanglement generation requirements). Then, on an n -hop path, the overall probability of successfully obtaining an E2E entangled pair, P_{E2E} , can be calculated as: $P_{E2E} = (p_{uv})^n \times (q_u)^{n-1}$. Notably, as the number of quantum repeaters increases, the success probability decreases exponentially [38]. Furthermore, it is crucial to emphasize that once entanglement is successfully generated, it exhibits a fleeting survival time due to its coherent interaction with the memory medium. Therefore, it is imperative to utilize the entanglement promptly to avoid long waiting periods. These probabilistic characteristics fundamentally distinguish QCNs from classical networks, necessitating distinct considerations in the design of routing and resource allocation schemes.

B. Motivation

This subsection provides a example to illustrate the necessity of our research. Taking Fig. 2(a) as a typical network,

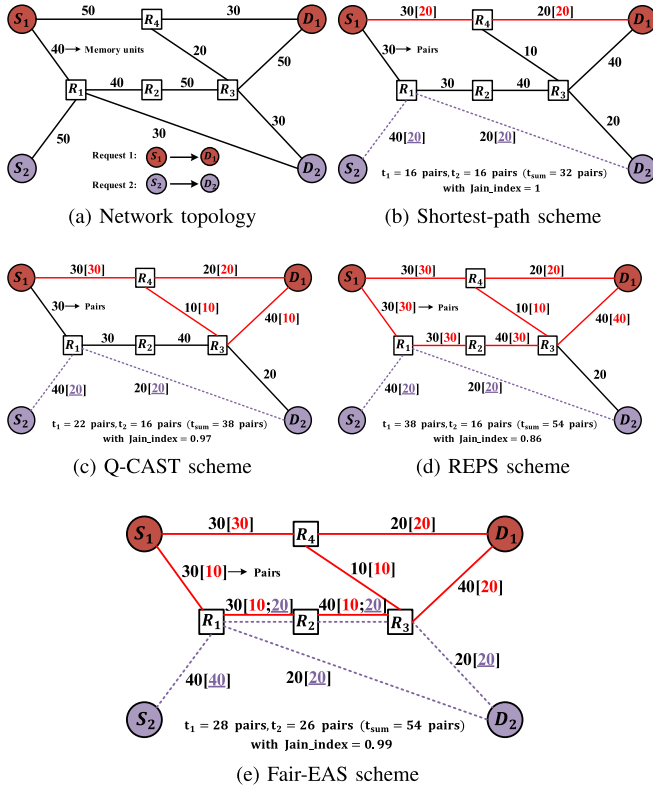


Fig. 2. (a) A typical network topology; (b-e) Four different E2E entanglement routing and resource allocation schemes.

there are two SD pairs in this network, (S_1, D_1) , (S_2, D_2) . The number on each link represents the maximum capacity of each link. In Fig. 2(b-e), solid links with numbers in the square bracket (no underline) are occupied for (S_1, D_1) , dotted links with numbers in the square bracket (underline) are occupied for (S_2, D_2) . The numbers in square brackets correspond to link width (i.e., the number of entangled pairs allocated on each link). In Fig. 2, t_1 and t_2 represent the expected number of E2E entangled pairs (i.e., expected throughput), refer to Eq. (1), while $t_{sum} = t_1 + t_2$. Jain's index of requests' expected throughput, abbreviated as $J_{req} = (t_1 + t_2)^2 / (2t_1^2 + 2t_2^2)$, shows the fairness between SD pairs (higher value represents better fairness). We set typical parameters $p_{uv} = q_u = 0.8$.

Although some simple routing strategies (for example, the shortest-path scheme [31], [32]) are easy to deploy and implement, it is evident that their throughput is not optimally considered. Expected distribution result by shortest path routing is only $t_1 = 16$ pairs, $t_2 = 16$ pairs as shown in Fig. 2(b). Despite Q-CAST [33] scheme can further improve the throughput, due to the limitation of distributed local link information exchange, the throughput improvement is not much (e.g., only recovery path $R_4 - R_3 - D_1$ can be found). For example, Q-CAST scheme distributes E2E entangled pairs as $t_1 = 22$ pairs, $t_2 = 16$ pairs in Fig. 2(c). The maximum throughput scheme REPS [10] exhibits expected distribution result as $t_1 = 38$ pairs, and $t_2 = 16$ pairs. However, it lacks good fairness between SD pairs, resulting in uneven resource allocation and noticeable throughput deviation with $J_{req} = 0.86$ in Fig. 2(d). Nevertheless, in Fig. 2(e), our Fair-EAS

scheme distributes E2E entangled pairs as $t_1 = 28$ pairs, and $t_2 = 26$ pairs with $J_{req} = 0.99$ meanwhile. Clearly, our Fair-EAS scheme maintains better fairness between the two SD pairs without sacrificing total throughput. Naturally, we aim to achieve similar fair routing and allocation results as Fair-EAS in any QCN, while maximizing the network throughput.

As demonstrated by the performance differences in the above cases, in addition to the limitations of hardware technology development, we believe that an important factor affecting the performance of E2E entanglement distribution is the result of routing and resource allocation. Specifically, path width, the number of hops in reserved paths, and the quality (success probability) of quantum operations. For example, a wider and shorter path (with more reserved resources and with fewer hops) always show better distribution efficiency and performance. However, we cannot simply allocate such "good" paths to all requests, which will inevitably affect the overall performance of the entire network and lack fairness considerations. Moreover, the method to ensure fairness is in QCNs is different from the average bandwidth allocation observed in classical networks. When formulating a new fair quantification model, the new challenges brought by the entanglement distribution process must be addressed, e.g., taking into account its probabilistic characteristics of generation and swapping. Finally, although the problem is complex, the corresponding solution should give priority to the simplicity of deployment and ensure compatibility with the inherent instability of entanglement.

IV. SYSTEM MODEL

A. Network Setting

1) *Centralized Controller and Time Slot*: Our research is still targeted at the primary development stage of QCNs, in which entanglement resources are still relatively scarce and the network scale is limited. Adopting a centralized routing architecture can better coordinate and schedule network resources and achieve higher resource utilization efficiency. Moreover, our research aims to achieve fairness towards the entanglement distribution process rather than the average allocation of local link resources. Distributed routing requires more interactions and computational iterations to coordinate resource competition between requests and meet the lower limit of requests' performance requirement. Then, to achieve cross-node entanglement distribution, all nodes on the path need to generate and hold entanglement with their predecessors and successors simultaneously. This necessitates time synchronization among all nodes, which can be achieved using existing synchronization protocols via classical networks. An appropriate synchronization duration ensures that the established entanglement remains stable within a single time slot [14], [33], [39]. Therefore, QCNs typically employ a time slot model to centrally control entanglement resources and handle multiple SD pairs. This model involves a global centralized controller that synchronizes with each node through classical channels, gathers information on entanglement resources, and schedules entanglement distribution tasks according to user requirements.

2) *Network Model*: Consider a QCN represented by a multigraph $\mathcal{G} = (V, E, C)$. Here, V denotes the set of nodes with the capability of quantum operations, E represents the set of edges in the graph, and the existence of an edge between two nodes implies that these nodes share one or more link-level entanglements. The set C encompasses all link-level entanglements, each identified by its two adjacent nodes, denoted as u and v . The maximum number of link-level entanglements on an edge (u, v) is referred to as the link capacity, denoted as $c(u, v)$.

B. Problem Definitions and Formulation

Different from widely used fair link bandwidth allocation, we design the expected throughput of a path as a metric to evaluate the fairness of the QCN. For a SD pair (S_i, D_i) , let $\mathcal{P}_i$ denote the set of all outputted paths between S_i and D_i , which contains m_i subpaths. Taking a subpath $P_j \in \mathcal{P}_i$, where $P_j = \{S_i = n_0, n_1, \dots, n_{l_j-1}, D_i = n_{l_j}\}$, l_j denotes the number of hops of P_j . Each link in path P_j assigns w_j memory units. Building an E2E entanglement connection successfully for an SD pair along a path requires all links to hold entanglements and all repeaters to swap successfully. The probability of one attempt to generate the entanglement successfully of all links and to swap successfully in all repeaters is $\prod_{x=1}^{l_j-1} q_{n_x} \prod_{x=1}^{l_j} p_{n_{x-1}n_x}$. Expected E2E entangled pairs t_i for SD pair i can be written as:

$$t_i = \sum_{j=1}^{m_i} \mathcal{D}(P_j) = \sum_{j=1}^{m_i} w_j \prod_{x=1}^{l_j-1} q_{n_x} \prod_{x=1}^{l_j} p_{n_{x-1}n_x}, \quad (1)$$

where $\mathcal{D}(P_j)$ represents the entanglement distribution process on P_j . Given the topology $\mathcal{G}$, as well as an SD pair set $\mathcal{S}$ composed of k users ($\mathcal{S} = \{(S_1, D_1), (S_2, D_2), \dots, (S_k, D_k)\}$), how can we determine the path set $\mathcal{P}_i$ for each SD pair i , and allocate entangled pairs on each subpath, so as to ultimately form the E2E throughput set $\mathcal{R} = \{t_1, t_2, \dots, t_k\}$ for all k users. We are committed to completing the following objectives:

$$\max_{\mathcal{R}} t_{\min} \quad (2)$$

$$s.t. \ t_i = \sum_{j=1}^{m_i} \mathcal{D}(P_j), \quad (2a)$$

$$t_i \geq t_{\min}, \forall t_i \in \mathcal{R}. \quad (2b)$$

$t_{\min}$ is the minimum component in $\mathcal{R}$. Obviously, $\mathcal{D}(P_j)$ in Eq. (1) is a nonlinear calculation, which leads to the problem becoming a difficult nonlinear optimization problem. Furthermore, considering the entanglement timeliness, how to decouple the problem reasonably, to solve quickly, and to implement results in QCNs is our focus in this study.

Set $\mathcal{R}$ is max-min fairness, if and only if for all t_i , $i \in \{1, \dots, k\}$, $t_i \geq t_{\min}$, while $t_{\min}$ cannot increase anymore. In other words, a system reaches max-min fairness, if it cannot increase any individual's resource without decreasing another individual's resource allocation which is already less than the previous ones. Hence, max-min fairness [40] proposed by Jaffe prevents any SD pair from "starvation", while increasing

the number of entangled pairs of each SD pair to a certain extent. For convenience, all the notations used in this paper are summarized in Table I.

V. FAIR-EAS SCHEME DESIGN

In this section, we elaborate the algorithm details of our Fair-EAS scheme in the offline and online phases, respectively. Finally, we provide assistance with complexity analysis and fairness proof.

A. Overview

In a globally synchronized time slot, similar to the existing methodologies, Fair-EAS systematically executes the following five phases for E2E entanglement distribution, ensuring generality and precision.

1. **Accept SD pairs**: The centralized controller collects the SD pairs sponsored by user nodes to form an SD pair set $\mathcal{S} = \{(S_i, D_i)\}$.
2. **Offline prediction**: The controller integrates the SD pair set information and the static offline topology of the network, completes multiple rounds of fairness optimization through our MWF algorithm (see Sec. V-B), and outputs the predicted path set for each SD pair.
3. **Link-level entanglement generation**: All links in the network complete the generation of link-level entangled pairs and notify the centralized controller of the global generation results.
4. **Online resource allocation**: Each SD pair allocates resources on its paths according to the ACE algorithm (see Sec. V-C) to compensate for the deviation of entangled pairs caused by probabilistic entanglement generation. Then, each SD pair reuses the entanglement "fragments" (remaining unallocated entangled pairs), which sporadically appear in the network by the EFE algorithm (see Sec. V-C).
5. **Distribute E2E entangled pairs**: According to the entangled pairs allocated in the previous phase, the controller tells the relevant repeaters to perform entanglement swapping to distribute E2E entangled pairs next.

Due to the fragility of entangled pairs, we deliberately divide the solution process into offline and online phases. The offline phase undertakes more complex pre-computation tasks and occurs before the network is operational, that is, before entanglement resource generation. In the online phase, time-saving resource allocation is made based on the prediction results to minimize the time overhead to mitigate decoherence. These two phases are the core of our study, influencing the resource allocation for each SD pair and thereby determining the ultimate outcome of entanglement distribution.

B. Offline Prediction

The most widely used algorithm for guaranteeing the max-min fairness is the water-filling (WF) algorithm, e.g., for the problem of communication bandwidth allocation. Based on the heuristics of the WF algorithm [41], [42], we propose multi-round water-filling-like optimization (MWF) algorithm aims to constantly find the smallest variable in

TABLE I
NOTATION LIST

Parameters	Description
$\mathcal{G} = (V, E, C)$	Network $\mathcal{G}$. V is the set of nodes, while E is the set of edges. C is the set of entanglements between adjacent nodes.
$\mathcal{S} = \{(S_i, D_i)\}$	Source-destination pair set.
$c(u, v)$	Link capacity of edge (u, v) .
$e_n(u, v)$	Equivalently transformed link capacity on edge (u, v) in n -round optimization based on Eq. (4).
p_{uv}, q_u	Entanglement generation probability on edge (u, v) ; Entanglement swapping probability on node u .
$f_i^n(u, v)$	Number of entanglements on edge (u, v) to be selected for SD pair i in n -round optimization.
t_i^n	Number of E2E entangled pairs for SD pair i in n -round optimization.
$\mathcal{T}_n$	Highest “water injection line” of n -round optimization.
$\mathcal{P}_i, m_i, l_j, w_j$	Path set $\mathcal{P}_i$ contains m_i subpaths for SD pair i . l_j and w_j denote the hops and width of subpath P_j .
E_i	The expected throughput of SD pair i .
A_i	The actual throughput of SD pair i .

$\mathcal{R} = \{t_i, t_2, \dots, t_k\}$ in turn. After each optimization, we always find the determined SD pair i with the smallest t_i , which is the current minimum component in $\mathcal{R}$. Then, we update the network topology $\mathcal{G}$ based on its corresponding routing results. The determined SD pair will be eliminated from $\mathcal{S}$ and the next round of optimization continues until $\mathcal{S}$ is empty. Specifically, we model the following multi-round optimization problem. In n -round, there exists:

n -round problem : $\max \mathcal{T}_n$ (3)

$$s.t. \sum_v f_i^n(u, v) - \sum_v f_i^n(v, u) = t_i^n, \quad \forall u \in \mathbb{S}, \quad (3a)$$

$$\sum_v f_i^n(u, v) - \sum_v f_i^n(v, u) = -t_i^n, \quad \forall u \in \mathbb{D}, \quad (3b)$$

$$\sum_v f_i^n(u, v) - \sum_v f_i^n(v, u) = 0, \quad \forall u \notin \mathbb{S}, \mathbb{D}, \quad (3c)$$

$$\sum_i^{k-n} f_i^n(u, v) + f_i^n(v, u) \leq e_n(u, v), \quad \forall (u, v) \in E, \quad (3d)$$

$$t_i^n \geq \mathcal{T}_n, i \leq k, \quad (3e)$$

$$\mathcal{T}_n \geq \mathcal{T}_{n-1} \geq 0, \quad (3f)$$

$$f_i^n(u, v) \in \mathbb{Z}^+. \quad (3g)$$

where $f_i^n(u, v)$ refers to the number of entangled pairs for SD pair i on the edge (u, v) . t_i^n represents the number of E2E entangled pairs for SD pair i . $\mathbb{S}$ and $\mathbb{D}$ are the source node set and the destination node set, respectively. The first three constraints, i.e., (3a)–(3c), are the flow conservation constraints that should be held in all the routing-related problems. Constraint (3d) reflects that the sum of all pending SD pairs on any link cannot exceed the current estimated number of link-level entangled pairs available $e_n(u, v)$. Especially, considering an entanglement distribution path, $\{S_0 = n_0, n_1, \dots, n_{l-1}, D_0 = n_l\}$, its successful probability can be calculated as $\prod_{i=1}^{l-1} q_{n_i} \prod_{i=1}^l p_{n_{i-1}n_i} = \prod_{i=1}^l p_{n_{i-1}n_i} \sqrt{q_{n_{i-1}}} \sqrt{q_{n_i}}$, if q_u is set to be 1 for $u = S_0$ or D_0 [10]. Thus, we slim the available link capacity equivalently

as follows:

$$e_0(u, v) = p_{uv} \sqrt{q_u} \sqrt{q_v} * c(u, v). \quad (4)$$

The equivalent conversion of the probability of swapping and generation to the modified probability of generation $p_{uv} \sqrt{q_u} \sqrt{q_v}$, as detailed in Eq. (4), transforms the non-linear entanglement distribution process into a linear one.

In each optimization round, we determine a maximum “water injection line” $\mathcal{T}_n$ to ensure that the expected throughput of each request t_i^n does not fall below this threshold, as specified in constraints (3e). Once the optimization problem is solved in n -round, the request whose expected throughput t_i^n is closest to $\mathcal{T}_n$ means its path set is determined. This request is considered the most disadvantaged in this round, indicating that further resource allocation to this request would compromise the performance of others. Additionally, we update the link capacity $e_n(u, v)$ according to the $f_i^n(u, v)$ allocated to the determined request i in this round. By repeating this process, we maintain that the “water injection line” for subsequent rounds does not decrease, i.e., $\mathcal{T}_n \geq \mathcal{T}_{n-1} \geq 0$, as stated in constraints (3f). This ensures that the expected throughput for determined requests in subsequent rounds is always higher or at least equal. The gradually increasing $\mathcal{T}_n$ realizes improving the global distribution performance on the premise of ensuring fairness. In essence, this process preferentially serves the most disadvantaged request and finishes its resource allocation while minimizing the adverse impact on the performance of other requests.

The integer variables in constraint (3g) render the optimization problem intractable for large-scale systems. Given the inevitable quantity deviation between offline predictive and the actual entanglement generation, we can reduce the complexity by sacrificing some accuracy in solving the optimal solution. Therefore, we relax the integer constraints on $f_i^n(u, v)$ and t_i^n , converting them into constraints on positive real numbers. This transforms the problem into a general linear programming (LP) problem, which can be efficiently solved using an optimization solver. Finally, the offline algorithm outputs the solved variable $f_i^n(u, v)$, which is the amount of entanglement resources allocated to request i on the relevant links. And these link resources are integrated and connected in series to form a path set. Moreover, at least one fairness individual (i.e., the routing result of one SD pair) is outputted after each optimization, so the circulation is limited, up to k times.

Algorithm 1 Multi-Round Water-Filling-Like Optimization (MWF) Algorithm

Input: Graph $\mathcal{G} = (V, E, C)$, SD pair set $\mathcal{S}$;

Output: Path set $\mathcal{P}'_i = \{P'_1, P'_2, \dots, P'_{m_i}\}$ for SD pair i ;

```

1  $e_0(u, v) = p_{uv} \sqrt{q_u} \sqrt{q_v} * c(u, v)$ ;
2 Slim network  $\mathcal{G}'_0$  composed of edge  $e_0(u, v)$ ;
3 Round  $n = 1$ .  $\mathcal{S}_0 = \mathcal{S}$ ;
4 while  $\mathcal{S}_{n-1} \neq \emptyset$  do
5   Relax the integer constraint and initialize
      $f'_i(u, v) = 0$ ;
6   Solve the relaxed optimization and derive the
     solutions  $f'_i(u, v)$  and  $t'_i$  for each SD pair;
7   Get  $\mathcal{P}'_i = \{P'_1, P'_2, \dots, P'_{m_i}\}$  for each SD pair;
8   Get residual graph  $\mathcal{G}'_n$  composed of the remaining
     resources;
9   for each SD pair  $i$  in  $\mathcal{S}_{n-1}$  do
10    if No path for SD pair  $i$  in  $\mathcal{G}'_n$  then
11      Remove determined SD pair  $i$  from  $\mathcal{S}_{n-1}$ ;
12      for each  $P'_j$  in  $\mathcal{P}'_i$  do
13        Get the path width  $w'_j$  of  $P'_j$ ;
14      end
15    end
16  end
17  Remove all determined  $\mathcal{P}'_i$  from  $\mathcal{G}'_{n-1}$  get  $\mathcal{G}'_n$ ;
18 end
19 Return Set  $\mathcal{P}'_i = \{P'_1, P'_2, \dots, P'_{m_i}\}$  for SD pair  $i$ .
```

We address this multi-round optimization problem as outlined in Alg. 1. The initialization in lines 1-3 focuses on calculating the equivalent link capacity $e_0(u, v)$ on each edge. Then, in lines 5-8, the integer constraint is relaxed first. Subsequently, the fractional path set $\mathcal{P}'_i$ for each SD pair i is obtained based on $f'_i(u, v)$ and t'_i . By removing the entanglement resources involved in the path set $\mathcal{P}'_i$ of each SD pair, we derive the network residual graph $\mathcal{G}'_n$. Lines 9-16 determine whether each SD pair is finalized in the n -round optimization by checking for the existence of additional paths in the residual graph $\mathcal{G}'_n$. If the search fails, it indicates that the routing calculation for this SD pair is complete, and the corresponding path set $\mathcal{P}'_i$ (comprising all subpaths and their corresponding path width w'_j) is available. Each round of optimization outputs at least one determined SD pair. Line 17 signifies the updating of the network graph based on the determined path set to obtain $\mathcal{G}'_n$ for the subsequent round.

C. Online Resource Allocation

After multiple rounds of offline optimization, each SD pair is assigned some exclusive paths, abbreviated as set $\mathcal{P}'_i$, with the expected number of entangled pairs w'_j allocated on each link along subpath P'_j . Despite considering the probabilities p_{uv} and q_u , the actual entanglement generation results remain dynamic and uncertain. Hence, the implementation of an adaptive compensatory resource allocation mechanism based on these actual outcomes is crucial to address potential extremes, such as insufficient or excessive entangled pairs.

Additionally, owing to the randomness of quantum operations and the optimization approximation, a small number of scattered entangled pairs may exist in the residual network, which are not accounted for in the preceding calculation. Consequently, the online resource allocation process primarily encompasses two key steps: an adaptive compensation algorithm of successfully generated entangled pairs and an entanglement “fragment” exhaustion algorithm.

1) *Adaptive Compensation Algorithm of Successfully Generated Entangled Pairs (ACE)*: Discrepancies between the offline multi-round optimization results and the actual network topology may arise, with some links having all entangled pairs generated successfully, while others exhibit fewer available entangled pairs than expected or none at all. These quantitative deviations, unaccounted for during the offline phase, necessitate adaptive adjustments to the entanglement allocation based on the actual generation results. Given the pre-existing expected allocation results from the offline phase, in consideration of decoherence and the need for expeditious correction, our objective is to design a rapid iterative adjustment algorithm to reduce time overhead—avoiding the intricacies of time-consuming optimization. The primary focus is on enhancing fairness among all SD pairs and establishing as many E2E entangled pairs as possible. If the actual number of entanglements generated falls short of the allocation result from offline optimization, conflicting links indicate a supply-demand mismatch. To address this, we propose the following two processing principles:

- **Fairness principle (P1)**: Based on the offline optimization results, the controller calculates the expected throughput $E_i = \sum_{j=1}^{m_i} w'_j$ of each SD pair i and sorts them in ascending order. The controller assigns the actual entanglement resources to the SD pair with minimum E_i first (giving priority to the inferior SD pair).
- **Throughput principle (P2)**: When allocating specific resources on each path in $\mathcal{P}'_i$ of SD pair i , priority should be given to shorter paths because resources on shorter paths are easier to form E2E entangled pairs with higher probability, which is conducive to improving throughput.

At the same time, w'_j represents the expected number of E2E entangled pairs instead of actual link resources. Therefore, we need to infer the number of link-level entangled pairs w_j that should be reserved on the path based on the probability of serial swapping. The reverse calculation is as follows:

$$w_j = \lfloor \frac{w'_j}{\prod_{x=1}^{l_j-1} q_{n_x}} \rfloor. \quad (5)$$

The actual allocation result for each SD pair should not exceed the path width w_j to avoid overcorrection. P1 reflects that SD pairs with lower expected throughput can be allocated resources preferentially. P2 ensures that shorter paths are allocated with priority in the path set of any SD pair which improves the throughput by reducing the times of entanglement swapping. Combining P1 and P2, the ACE algorithm can assign an appropriate number of entangled pairs on each SD pair’s multiple paths in turn, without exceeding the upper bound generated by offline optimization.

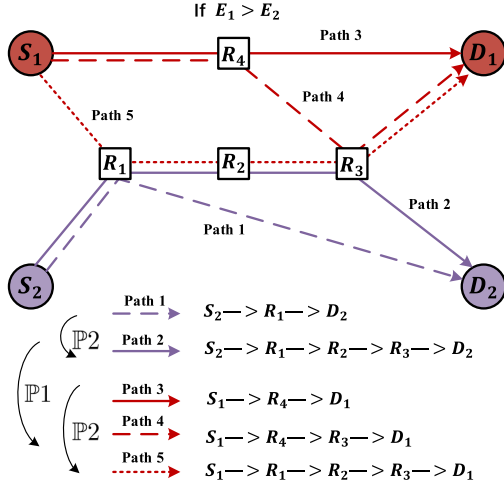


Fig. 3. Online ACE algorithm allocates the actual entanglement resources on the determined path set according to $\mathbb{P}1$ and $\mathbb{P}2$.

A typical case of the ACE algorithm, depicted in Fig. 3, illustrates how to determine the online resource allocation sequence for the alternative path set in accordance with the core principles $\mathbb{P}1$ and $\mathbb{P}2$. If $E_1 > E_2$, (S_2, D_2) is identified as the less dominant one between the two SD pairs. According to the fairness principle $\mathbb{P}1$, the ACE algorithm prioritizes resource allocation for (S_2, D_2) , followed by (S_1, D_1) , based on the offline calculation results. When allocating resources to multiple paths for each request, we adopt the throughput principle $\mathbb{P}2$ to prioritize serving the paths with fewer hops. This prioritization is manifested as: path1 $\rightarrow$ path2 for (S_2, D_2) ; and path3 $\rightarrow$ path4 $\rightarrow$ path5 for (S_1, D_1) . If we implement the resource allocation process in reverse, the performance gap of entanglement distribution between the two requests would not be reduced, and the effect of online resource allocation compensation would not be achieved.

The details of the ACE algorithm are shown in Alg. 2. First, all the basic links in the network perform entanglement generation and inform results to the centralized controller. Then the network controller forms the actual entanglement generation graph $\hat{G}$. At the same time, the controller also masters the optimal path set $\mathcal{P}'_i$ output by offline optimization. Next, in lines 1-5, we calculate the expected throughput of each SD pair based on the optimization results. Lines 6-14 explain that the priority of allocating resources should be given to SD pairs of smaller E_i according to $\mathbb{P}1$, and the allocation result should be the smaller value of the optimization result w_j and the actual resources w_{min} . Then, the shorter path in $\mathcal{P}'_i$ of one SD pair should be allocated preferentially to improve throughput by turn, i.e., $\mathbb{P}2$. Finally, the ACE algorithm quickly adjusts the allocation results according to the real-time resources and outputs the validated and revised path set $\hat{\mathcal{P}}_i$ of each SD pair.

2) *Entanglement "Fragment" Exhaustion (EFE) Algorithm:* Following the previous offline optimization and online resource allocation, residual entangled pairs, unclaimed by any SD pair, are dispersed across the network, forming entanglement "fragments". They are shared resources, accessible to any SD pair. To enhance network throughput and resource

Algorithm 2 Adaptive Compensation Algorithm of Successfully Generated Entangled Pairs (ACE)

Input: Actual generated graph $\hat{G} = (\hat{V}, \hat{E}, \hat{C})$; path set $\mathcal{P}'_i = \{P'_1, P'_2, \dots, P'_{m_i}\}$ for SD pair i ; SD pair set $\mathcal{S}$;
Output: Actual path set $\hat{\mathcal{P}}_i = \{\hat{P}_1, \hat{P}_2, \dots, \hat{P}_{m_i}\}$ for each SD pair i ;

```

1 for each SD pair  $i$  in  $\mathcal{S}$  do
2    $\mathcal{P}'_i = \{P'_1, P'_2, \dots, P'_{m_i}\} \leftarrow \text{Algorithm 1}$ ;
3   The expected throughput  $E_i = \sum_{j=1}^{m_i} w'_j$ ;
4 end
5 Sort SD pairs of  $\mathcal{S}$  in ascending order by  $E_i$ ;
6 for each SD pair  $i$  in  $\mathcal{S}$  do
7   Sort all subpaths in ascending order by hops;
8   for each  $P'_j$  in  $\mathcal{P}'_i$  do
9     Get actual width  $w_j = \lfloor \frac{w'_j}{\prod_{x=1}^{l_j-1} q_{n_x}} \rfloor$ ;
10    Actual minimum link width  $w_{min}$  on  $P'_j$ ;
11    Actual allocatable width  $\hat{w}_j = \min(w_{min}, w_j)$ 
    and corresponding path  $\hat{P}_j$ ;
12    Remove  $\hat{P}_j$  from  $\hat{G}$ ;
13  end
14 end
15 Return Set  $\hat{\mathcal{P}}_i = \{\hat{P}_1, \hat{P}_2, \dots, \hat{P}_{m_i}\}$  for each SD pair  $i$ .
```

utilization, we aim to consolidate these "fragments" to create additional E2E entangled pairs. The subsequent emphasis is on refining the allocation strategy for these residual entangled pairs among all SD pairs.

Specifically, the EFE algorithm employs a priority-based minimum granularity resource allocation approach, where the priority is determined by the current actual throughput $A_i = \sum_{j=1}^{m_i} \hat{w}_j \prod_{x=1}^{l_j-1} q_{n_x}$ of each request ($\hat{w}_j$ means the number of entangled pairs reserved on each link along the subpath $\hat{P}_j$). In each iteration, SD pairs exhibiting lower actual throughput A_i prioritize to search a residual path, aiming to enhance the distribution performance of non-dominant requests. The principle of minimum granularity ensures that only a single entangled pair is utilized on each link of the residual path, preventing any SD pair from monopolizing excessive resources. Furthermore, upon discovering a residual path, the algorithm promptly updates the network topology and the path set associated with each request, initiating the next round of residual path exploration. Upon completion of the EFE algorithm, it becomes evident that there is no further room for optimizing the throughput within the current network.

As shown in Fig. 4, there is a residual graph composed of entanglement "fragments", which contains only a few remaining entangled pairs (represented by the black solid lines). We can determine how to connect these "fragments" by calculating the actual throughputs of the two requests. If the actual throughput of two SD pairs is $A_1 < A_2$, the EFE algorithm uses five pairs on links $S_1 - R_1$, $R_1 - R_2$, $R_2 - R_3$, $R_3 - R_4$, and $R_4 - D_1$ to distribute the entangled pair for (S_1, D_1) . Conversely, if the actual throughput of two SD pairs is $A_1 \geq A_2$, the EFE algorithm arranges four pairs on links

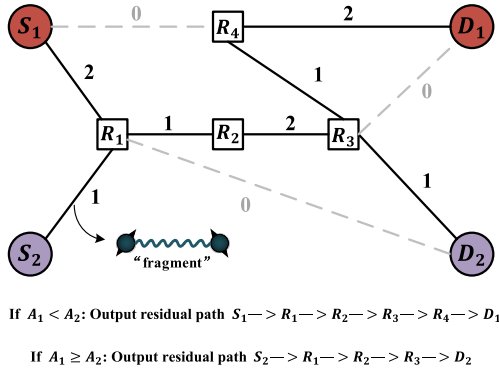


Fig. 4. Typical instance of entanglement “fragment” exhaustion algorithm in a sample residual graph.

$S_2 - R_1$, $R_1 - R_2$, $R_2 - R_3$, and $R_3 - D_2$ to distribute the entangled pair for (S_2, D_2) . Specifically, when $A_1 = A_2$, since there are fewer hops on the path between (S_2, D_2) , we still choose to connect the above four pairs in series because of the higher distribution success probability.

Algorithm 3 Entanglement “fragment” Exhaustion (EFE) Algorithm

Input: Residual Graph $\underline{G} = (\underline{V}, \underline{E}, \underline{C})$; path set $\hat{P}_i = \{\hat{P}_1, \hat{P}_2, \dots, \hat{P}_{m_i}\}$ for SD pair i ; SD pair set $\mathcal{S}$;

Output: Final path set $\hat{P}_i = \{\hat{P}_1, \hat{P}_2, \dots, \hat{P}_{s_i}\}$ for SD pair i , and $s_i \geq m_i$;

- 1 Collecting entanglement “fragments” to form $\underline{G}$;
- 2 flag = 1; /* Entanglement fragments are not fully utilized. */
- 3 **while** flag **do**
- 4 **for each** SD pair i in $\mathcal{S}$ **do**
- 5 The actual throughput $A_i = \sum_{j=1}^{m_i} \hat{w}_j \prod_{x=1}^{l_j-1} q_{n_x}$;
- 6 $\leftarrow$ **Algorithm 2**;
- 7 **end**
- 8 Sort SD pairs of $\mathcal{S}$ in ascending order by A_i ;
- 9 **for each** SD pair i in $\mathcal{S}$ **do**
- 10 **if** find a path $\underline{P}_i$ for SD pair i in $\underline{G}$ **then**
- 11 Add path $\underline{P}_i$ with $w = 1$ to $\hat{P}_i$;
- 12 Remove path $\underline{P}_i$ from $\underline{G}$;
- 13 **break**;
- 14 **end**
- 15 flag = 0;
- 16 **end**
- 17 **Return** Final set $\hat{P}_i = \{\hat{P}_1, \hat{P}_2, \dots, \hat{P}_{s_i}\}$ for SD pair i .

Next, we will describe the exhaustive details of the EFE algorithm in Alg. 3. First, in lines 3-7, we calculate the actual throughput A_i of each SD pair i according to the resource allocation result of the ACE algorithm and sort them in ascending order. Lines 8-16 reflect the EFE algorithm tries to find one path $\underline{P}_i$ for the SD pair i with current minimum A_i . The controller can be applied to some efficient routing

algorithms, such as Dijkstra. If a path $\underline{P}_i$ is found for the current SD pair i , an entangled pair is reserved on each link of this path $\underline{P}_i$, i.e., a path with path width $w = 1$. On the contrary, if the search fails, the algorithm continues to search for the next SD pair. The controller then updates the actual throughput A_i of each SD pair and sorts them in ascending order. The controller repeats the pathfinding and resource-locking steps until no path can be found for any SD pair.

D. Algorithm Analysis

1) *Complexity Analysis of the MWF Algorithm:* The MWF algorithm entails multiple rounds of iterative linear programming optimization, potentially up to k times. Each iteration involves the solution of a linear programming problem, and in the case of employing the Interior Point Method, the worst-case time complexity is expressed as $O(n^3 L)$ [43], where n represents the number of decision variables, and L is a parameter reflecting the size of the problem data. It is crucial to emphasize that the count of decision variables in each round remains within the limit of $k|E|$, where $|E|$ denotes the total number of links. Consequently, the upper bound on the overall complexity of WMF is established as $O(kn^3 L) = O(k^4 |E|^3 L)$.

2) *Complexity Analysis of ACE Algorithm and EFE Algorithm:* The ACE algorithm necessitates traversing the actual number of entangled pairs along the selected path. The traversal complexity is linear with the amount of data. Thus, for all SD pairs, the time complexity of ACE is denoted as $O(\sum_{i=1}^k \sum_{j=1}^{m_i} H_j)$, where H_j represents the hops of subpath j , and m_i denotes the number of subpaths for SD pair i . The maximum number ($\sum_{j=1}^{m_i} H_j$) of links occupied by each SD pair i does not exceed all links ($|E|$) in the entire network. Consequently, the upper bound of the time complexity of ACE is established as $O(k|E|)$.

The EFE algorithm employs the shortest path algorithm to identify potential remaining paths for each SD pair on a sparse residual graph, specifically Dijkstra, with time complexity of $O(|V|^2)$, where $|V|$ denotes the number of nodes in the residual graph. Assuming that the average number of discoverable paths is m , the algorithm’s overall complexity is expressed as $O(km|V|^2)$. Subsequent simulations have revealed that in practice, the value of m tends to be small, often approaching or equal to zero. In summary, the design of two low-complexity online algorithms is effective in dealing with the short timeliness of entanglement.

Lemma 1: The MWF algorithm always concludes within no more than k rounds of optimization. Then, when $\mathcal{S}_n = \emptyset$, the set $\mathcal{R}$ satisfies the optimal max-min fairness.

Proof: Given that the entanglement resources set in graph $\mathcal{G}'_n$ in Alg. 1 is both finite and compact, it implies that the maximum value $\mathcal{T}_n$ in the n -round optimization can be attained. If $\mathcal{T}_n$ exists, and for all $i \in \mathcal{S}_n$ but $i \notin \mathcal{S}_{n-1}$ i.e., $i \in \mathcal{S}_n \setminus \mathcal{S}_{n-1}$, there exists $t_i \geq \mathcal{T}_n$. Consequently, $\mathcal{S}_n \subset \mathcal{S}_{n-1}$ holds. This leads to the conclusion that the cardinality $|\mathcal{S}_n|$ decreases, eventually reaching $\mathcal{S}_n = \emptyset$ in at most k steps. Subsequently, if we arrange the set $\mathcal{R}$ in ascending order,

i.e., $\mathcal{R} = \{t_1, t_2, \dots, t_k\}$. According to constraints (3f), this ordering in set $\mathcal{R}$ also corresponds to the solving order of Alg. 1. In the initial round of solving, Eq. (3) maximizes t_1 while ensuring $t_1 \leq \mathcal{R} \setminus \{t_1\}$. This process repeats, and after the i -th round of solving, we attain the maximum t_i that satisfies $t_{i-1} \leq t_i \leq \mathcal{R} \setminus \{t_1, \dots, t_{i-1}\}$. Consequently, upon completion of the algorithm, the output results adhere to $t_1 \leq t_2 \leq \dots \leq t_k$, and it ensures that at position i in the set $\mathcal{R}$, t_i is the maximum value that can be taken. Therefore, when $\mathcal{S}_n = \emptyset$, the final set $\mathcal{R}$ satisfies max-min fairness. ■

VI. PERFORMANCE EVALUATION

In this section, we simulate the performance of our Fair-EAS scheme and the typical comparison schemes in various scenarios. We then evaluate and analyze the simulation results in detail comprehensively.

A. Methodology

Our simulation code is developed based on Python 3.10. The LP solver used in our simulator is PuLP,¹ which is an open-source third-party toolkit that can solve linear programming problems. The key setting information and evaluation methods used in the simulation are listed below:

1) *Simulation Setting*: In our simulation, we introduce the 50-node Waxman random topology. Edges of random network graphs are determined following the Waxman model [44], in which we set parameter $\alpha = 0.25$, $\beta = 0.25$. The link capacity of the random network graph is $c(u, v) = 10$, signifying that after the entanglement generation phase, the actual number of external connections is not more than 10. Then the locations of SD pairs in network are random with no repetition, i.e., the SD pair can be composed of two nodes at any location. For each result, we run 100 times to minimize errors and eliminate contingencies. In addition to calculating the average of various performance values, we incorporate standard deviation to emphasize randomness.

2) *Comparison Schemes*: Considering the consistency of the system model, we compare Fair-EAS with two representative entanglement routing schemes. One is Q-CAST scheme which provides the first efficient routing scheme to establish the entangled pairs. The path recovery algorithm in Q-CAST scheme is based on the exchange of local link state information. In order to facilitate comparison with our centralized scheme, we use global link information to improve the path recovery mechanism, called Q-CAST+, which obviously outperforms the original Q-CAST scheme. The other is REPS scheme as it is shown to have the highest throughput so far through network flow optimization.

3) *Performance Metrics*: We design the following four metrics to evaluate performance.

- **Total throughput**: Throughput sum of all k SD pairs $\sum_i^k t_i$, where t_i is the E2E throughput of SD pair i . This metric can reflect the total service capacity of the routing scheme for all SD pairs.
- **Minimum throughput**: Minimum throughput of all SD pairs. It can show the performance guarantee for

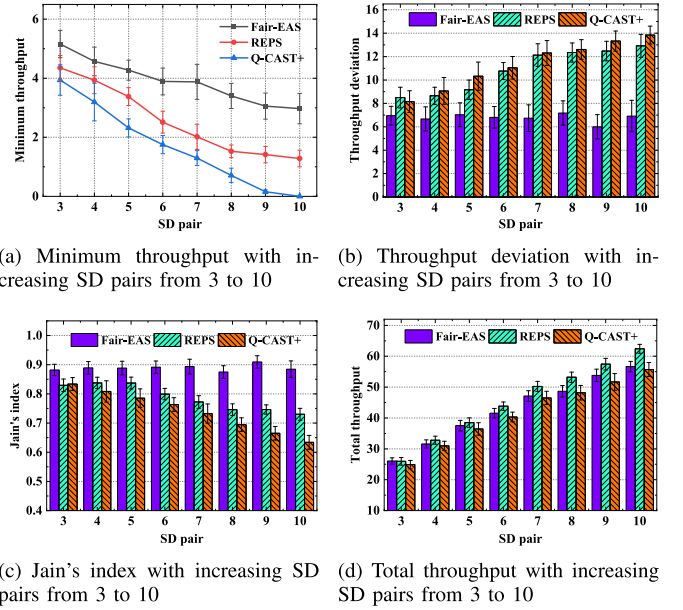


Fig. 5. Simulation results of increasing SD pairs in a randomly generated network with 50 nodes.

not-dominant SD pairs between different schemes, i.e., max-min fairness.

- **Jain's index**: Jain's index of requests' throughput, abbreviated as J_{req} [45], is adopted to evaluate the fairness of the different routing algorithms:

$$J_{req} = \frac{(\sum_i^k t_i)^2}{k(\sum_i^k t_i^2)}, \quad (6)$$

where $J_{req} \in [0, 1]$, and $J_{req} = 1$ represents complete fairness due to the same throughput of each request.

- **Throughput deviation**: The maximum throughput minus the minimum throughput to evaluate the range between requests' throughput. Smaller deviation indicates more close performance between requests.

B. Simulation Results

1) *Simulation With Increasing SD Pairs*: As illustrated in Fig. 5, the four subfigures present a comparison of minimum throughput, throughput deviation, Jain's index, and total throughput with the increasing SD pairs in the random topology network, respectively. In this simulation, the positions of SD pairs are randomly assigned, and we set $p_{uv} = 0.8$ and $q_u = 0.8$.

Fig. 5(a) demonstrates that Fair-EAS scheme ensures a higher minimum throughput, thereby enhancing the lower bound of entanglement distribution performance for any SD pairs. Notably, the improvement in $t_{\min}$ is often several times greater with an increasing number of SD pairs. Even when deploying 9 or 10 SD pairs, for REPS and Q-CAST+ schemes, the minimum throughput becomes very small or even approaches zero. This implies that at least one SD pair fails to form a valid E2E entangled pair. Fig. 5(b) reveals that the throughput deviation of Fair-EAS remains the lowest. Moreover, Fair-EAS scheme does not exhibit an increasing trend as the number of SD pairs grows, indicating a better

¹<https://coin-or.github.io/pulp/>

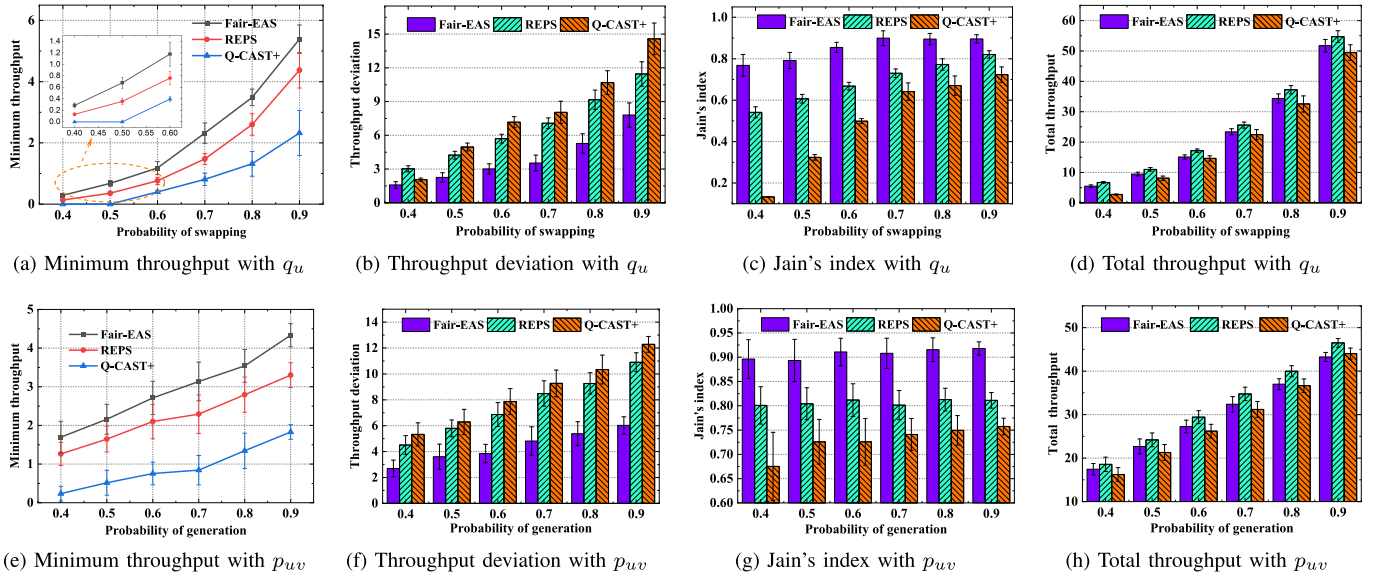


Fig. 6. Simulation results of 6 SD pairs with the increasing q_u (a-d) and p_{uv} (e-h) in a randomly generated network with 50 nodes.

balance among SD pairs. In contrast, the throughput deviation of the other two schemes consistently shows an upward trend, signifying that certain SD pairs (located closely with few hops) attain very high throughput, while others (positioned remotely with many hops) experience poor throughput. This drawback becomes more pronounced as the number of SD pairs increases.

In Fig. 5(c), the Jain's index of the three schemes is presented with the increasing SD pairs. Our Fair-EAS scheme achieves a fairness guarantee with Jain's index close to 1, and this attenuation is minimal as the number of SD pairs increases (remaining almost unchanged). In contrast, REPS and Q-CAST+ schemes exhibit notably poor fairness. Their Jain's index decreases rapidly as the number of SD pairs increases, and even in the case of 10 SD pairs, their Jain's index is only half of that achieved by Fair-EAS scheme. A Jain's index of 0.5 reflects very poor fairness, as Jain's index = 0.1 indicates total injustice. Fig. 5(d) illustrates the total throughput of the three schemes with the increasing SD pairs. Our Fair-EAS scheme closely matches or surpasses the throughput of REPS and Q-CAST+ schemes, which optimize the overall network throughput when there are not many SD pairs. Although Fair-EAS scheme exhibits a slight disadvantage as the number of SD pairs increases, the decline in throughput is a normal side effect of ensuring fairness.

2) *Simulation With Increasing p_{uv} and q_u* : As depicted in Fig. 6, the eight subfigures provide a comparative analysis of minimum throughput, throughput deviation, Jain's index, and total throughput with the increasing operation probability. The variations in the probability of entanglement swapping q_u or entanglement generation p_{uv} from 0.4 to 0.9 are explored while keeping another probability fixed at 0.8. The simulation involves deploying 6 SD pairs at 12 random locations within a 50-node random topology network. In this context, the Fair-EAS algorithm consistently demonstrates significant superiority with higher minimum throughput and narrower deviation over the other two algorithms, as shown

in Fig. 6(a-b) and Fig. 6(e-f). Specifically, the increasing probability p_{uv} of entanglement generation corresponds to an expansion in network resources. Consequently, the potential number of available entangled pairs for each SD pair gradually increases. Subsequently, the increasing probability q_u of entanglement swapping results in higher-quality operations for easier entanglement distribution. Both of them lead to improvement in both minimum throughput and deviation. However, for Q-CAST+ algorithm, when the swapping probability is very low, certain requests may not even receive a sufficient amount of entangled pairs to distribute an effective entangled pair.

In our approach, the offline prediction considers probabilities of entanglement generation and swapping. In Fig. 6(c) and Fig. 6(g), an increase in these probabilities improves the accuracy of offline predictions, leading to higher Jain's index and more stable distribution results. Conversely, REPS and Q-CAST+ lack full integration of these quantum properties in the entanglement distribution process, resulting in uneven resource allocation and a detrimental impact on overall performance. As probability increases and uncertainty weakens, entanglement distribution becomes more similar to classical processes, so the gap between these algorithms is also decreasing. Observing throughput in Fig. 6(d) and Fig. 6(h), when the probability of quantum operation is low, that is, quantum characteristics are more obvious. Our Fair-EAS scheme can maintain a total throughput close to the optimal level. Even for high operation probability, due to the multiple mechanisms designed to improve throughput, a good balance between throughput and fairness is maintained.

C. Discussion

1) *Robustness*: To improve robustness for mitigating the impact of quantum decoherence, efforts in these aspects can be further considered: (1) As the network scale expands, the network can be divided into domains [29] for management,

or a recursive network structure [46] can be adopted. Then, end-to-end entanglement distribution can be decomposed into several entanglement distribution tasks within domains. The number of network nodes under centralized control in each domain decreases, and the computational time consumption of the algorithm will also be reduced. (2) Researchers develop quantum memories with higher storage efficiency and longer storage time as soon as possible to enhance the resistance ability of entangled pairs against quantum decoherence. (3) Using redundant resources for entanglement purification can improve the quality of entangled pairs and prolong their effective lifetime.

2) *Scalability*: From a theoretical perspective, as shown in the complexity analysis in Sec. V-D, as the network scale expands and the number of requests increases, our offline algorithm will unavoidably lead to significant increase in computational time consumption due to the more decision variables in the linear programming problem. However, since this occurs before entanglement generation, it remains insensitive to the computational time consumption of multi-round solutions.

Additionally, the time complexity of the online algorithm remains at a low-order polynomial level. Currently, QCN is still in its primary stage, with limited network scale and requests. Our online resource allocation algorithm based on synchronous control still has efficient and reliable deployment value. Even if the network scale expands, managers can divide the network into domains for management. The quantum local area network (QLAN) adopting domain-based autonomy can greatly reduce the problem complexity of entanglement routing and resource allocation, which can improve the real-time performance of each request.

VII. CONCLUSION AND FUTURE WORK

In this study, we introduced Fair-EAS, an innovative approach designed to establish a process-oriented fairness guarantee for multiple requests in QCNs. In contrast to the existing work, we modeled max-min fairness by utilizing expected throughput, and presented a novel offline-online step-wise solution tailored to entanglement properties, including the probability of operations and decoherence. Simulation results and analysis demonstrated that our Fair-EAS scheme significantly enhances the minimum throughput and mitigates the deviation of throughput between multiple SD pairs. Concurrently, Fair-EAS maintains a nearly optimal overall network throughput and achieves a balance between fairness guarantee and improving network performance. Our research unveils the potential for valuable process-oriented fair entanglement allocation and selection in QCNs, promising better deployability and the integration of novel functionalities.

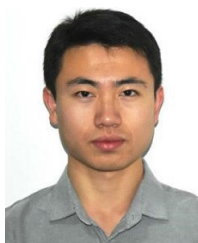
Based on this study, we can consider these potential follow-up research. For example, we can apply software-defined networking (SDN) technology to manage and monitor the entire network more efficiently, and achieve more efficient resource allocation and synchronization function, especially in large-scale quantum networks with domain-based governance. Moreover, we can introduce the entanglement purification mechanism and consider a routing and resource allocation

scheme coupled with purification, that is, an evolved version of this research, to resist quantum decoherence and improve real-time performance.

REFERENCES

- [1] W. Kozłowski, A. Dahlberg, and S. Wehner, "Designing a quantum network protocol," in *Proc. Int. Conf. Emerg. Netw. Exp. Technol. (CoNEXT)*, 2020, pp. 1–16.
- [2] H.-K. Lo, M. Curty, and K. Tamaki, "Secure quantum key distribution," *Nature Photon.*, vol. 8, no. 8, pp. 595–604, 2014.
- [3] P. Komar et al., "A quantum network of clocks," *Nature Phys.*, vol. 10, no. 8, pp. 582–587, 2014.
- [4] E. O. Ilo-Okeke, L. Tessler, J. P. Dowling, and T. Byrnes, "Remote quantum clock synchronization without synchronized clocks," *NPJ Quantum Inf.*, vol. 4, no. 1, p. 40, Aug. 2018.
- [5] J. F. Fitzsimons and E. Kashefi, "Unconditionally verifiable blind quantum computation," *Phys. Rev. A, Gen. Phys.*, vol. 96, no. 1, Jul. 2017, Art. no. 012303.
- [6] H. Riel, "Quantum computing technology," in *IEDM Tech. Dig.*, Dec. 2021, pp. 1–3.
- [7] L. Chen et al., "A heuristic remote entanglement distribution algorithm on memory-limited quantum paths," *IEEE Trans. Commun.*, vol. 70, no. 11, pp. 7491–7504, Nov. 2022.
- [8] Z. Li et al., "NarrowGap: Reducing bottlenecks for end-to-end entanglement distribution in quantum networks," *IEEE/ACM Trans. Netw.*, early access, Oct. 16, 2024, doi: 10.1109/TNET.2024.3476342.
- [9] Z. Xiao et al., "Purification scheduling control for throughput maximization in quantum networks," *Commun. Phys.*, vol. 7, no. 1, p. 307, Sep. 2024.
- [10] Y. Zhao and C. Qiao, "Redundant entanglement provisioning and selection for throughput maximization in quantum networks," in *Proc. IEEE Conf. Comput. Commun. (INFOCOM)*, May 2021, pp. 1–10.
- [11] L. Chen et al., "Q-DDCA: Decentralized dynamic congestion avoid routing in large-scale quantum networks," *IEEE/ACM Trans. Netw.*, vol. 32, no. 1, pp. 368–381, Jan. 2023.
- [12] Z. Li et al., "Swapping-based entanglement routing design for congestion mitigation in quantum networks," *IEEE Trans. Netw. Service Manage.*, vol. 20, no. 4, pp. 3999–4012, Apr. 2023.
- [13] C. Li, T. Li, Y.-X. Liu, and P. Cappellaro, "Effective routing design for remote entanglement generation on quantum networks," *NPJ Quantum Inf.*, vol. 7, no. 1, pp. 1–12, Jan. 2021.
- [14] C. Cicconetti, M. Conti, and A. Passarella, "Request scheduling in quantum networks," *IEEE Trans. Quantum Eng.*, vol. 2, pp. 2–17, 2021.
- [15] J. Li et al., "Fidelity-guaranteed entanglement routing in quantum networks," *IEEE Trans. Commun.*, vol. 70, no. 10, pp. 6748–6763, Oct. 2022.
- [16] A. S. Cacciapuoti, M. Caleffi, F. Tafuri, F. S. Cataliotti, S. Gherardini, and G. Bianchi, "Quantum Internet: Networking challenges in distributed quantum computing," *IEEE Netw.*, vol. 34, no. 1, pp. 137–143, Jan. 2019.
- [17] J. Shi and S. Shen, "A clock synchronization method based on quantum entanglement," *Sci. Rep.*, vol. 12, no. 1, p. 10185, Jun. 2022.
- [18] S. Wehner, D. Elkouss, and R. Hanson, "Quantum Internet: A vision for the road ahead," *Science*, vol. 362, no. 6412, p. 9288, Oct. 2018.
- [19] W. Dai, T. Peng, and M. Z. Win, "Optimal remote entanglement distribution," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 3, pp. 540–556, Mar. 2020.
- [20] H.-J. Briegel, W. Dür, J. I. Cirac, and P. Zoller, "Quantum repeaters: The role of imperfect local operations in quantum communication," *Phys. Rev. Lett.*, vol. 81, p. 5932, Dec. 1998.
- [21] L.-M. Duan, M. D. Lukin, J. I. Cirac, and P. Zoller, "Long-distance quantum communication with atomic ensembles and linear optics," *Nature*, vol. 414, no. 6862, pp. 413–418, 2001.
- [22] N. Sangouard, C. Simon, H. De Riedmatten, and N. Gisin, "Quantum repeaters based on atomic ensembles and linear optics," *Rev. Mod. Phys.*, vol. 83, no. 1, p. 33, 2011.
- [23] R. Mazumdar, L. Mason, and C. Douligeris, "Fairness in network optimal flow control: Optimality of product forms," *IEEE Trans. Commun.*, vol. 39, no. 5, pp. 775–782, May 1991.
- [24] A. Dahlberg et al., "A link layer protocol for quantum networks," in *Proc. Annu. Conf. ACM Special Interest Group Data Commun. Appl., Technol., Archit., Protocols Comput. Commun. (SIGCOMM)*, 2019, pp. 159–173.

- [25] Y. Lei, F. Kimiaee Asadi, T. Zhong, A. Kuzmich, C. Simon, and M. Hosseini, "Quantum optical memory for entanglement distribution," *Optica*, vol. 10, no. 11, pp. 1511–1528, 2023.
- [26] X.-J. Wang et al., "Cavity-enhanced atom-photon entanglement with subsecond lifetime," *Phys. Rev. Lett.*, vol. 126, no. 9, Mar. 2021, Art. no. 090501.
- [27] N. Bar-Gill, L. M. Pham, A. Jarmola, D. Budker, and R. L. Walsworth, "Solid-state electronic spin coherence time approaching one second," *Nature Commun.*, vol. 4, no. 1, p. 1743, Apr. 2013.
- [28] Z. Li et al., "Entanglement-assisted quantum networks: Mechanics, enabling technologies, challenges, and research directions," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 4, pp. 2133–2189, 4th Quart., 2023.
- [29] Z. Li et al., "Building a large-scale and wide-area quantum internet based on an OSI-alike model," *China Commun.*, vol. 18, no. 10, pp. 1–14, Oct. 2021.
- [30] L. Chen et al., "REDP: Reliable entanglement distribution protocol design for large-scale quantum networks," *IEEE J. Sel. Areas Commun.*, vol. 42, no. 7, pp. 1723–1737, Jul. 2024.
- [31] R. Van Meter, T. Satoh, T. D. Ladd, W. J. Munro, and K. Nemoto, "Path selection for quantum repeater networks," *Netw. Sci.*, vol. 3, pp. 82–95, Dec. 2013.
- [32] M. Pant et al., "Routing entanglement in the quantum internet," *NPJ Quantum Inf.*, vol. 5, no. 1, pp. 1–9, Mar. 2019.
- [33] S. Shi and C. Qian, "Concurrent entanglement routing for quantum networks: Model and designs," in *Proc. Annu. Conf. ACM Special Interest Group Data Commun. Appl., Technol., Archit., Protocols Comput. Commun.*, 2020, pp. 62–75.
- [34] R. Van Meter and J. Touch, "Designing quantum repeater networks," *IEEE Commun. Mag.*, vol. 51, no. 8, pp. 64–71, Aug. 2013.
- [35] Z. Li, K. Xue, J. Li, N. Yu, D. S. L. Wei, and R. Li, "Connection-oriented and connectionless remote entanglement distribution strategies in quantum networks," *IEEE Netw.*, vol. 36, no. 6, pp. 150–156, Nov. 2022.
- [36] J. Yin et al., "Satellite-based entanglement distribution over 1200 kilometers," *Science*, vol. 356, no. 6343, pp. 1140–1144, Jun. 2017.
- [37] D. Chandra, A. S. Cacciapuoti, M. Caleffi, and L. Hanzo, "Direct quantum communications in the presence of realistic noisy entanglement," *IEEE Trans. Commun.*, vol. 70, no. 1, pp. 469–484, Jan. 2022.
- [38] Z. Wang et al., "An asynchronous entanglement distribution protocol for quantum networks," *IEEE Netw.*, vol. 36, no. 5, pp. 40–47, Sep. 2022.
- [39] K. Chakraborty, D. Elkouss, B. Rijsman, and S. Wehner, "Entanglement distribution in a quantum network: A multicommodity flow-based approach," *IEEE Trans. Quantum Eng.*, vol. 1, pp. 1–21, 2020.
- [40] B. Radunovic and J.-Y. Le Boudec, "A unified framework for max-min and min-max fairness with applications," *IEEE/ACM Trans. Netw.*, vol. 15, no. 5, pp. 1073–1083, 2007.
- [41] W. Yu, W. Rhee, S. Boyd, and J. M. Cioffi, "Iterative water-filling for Gaussian vector multiple-access channels," *IEEE Trans. Inf. Theory*, vol. 50, no. 1, pp. 145–152, Jan. 2004.
- [42] D. P. Palomar and J. R. Fonollosa, "Practical algorithms for a family of waterfilling solutions," *IEEE Trans. Signal Process.*, vol. 53, no. 2, pp. 686–695, Feb. 2005.
- [43] D. D. Hertog, *Interior Point Approach to Linear, Quadratic and Convex Programming: Algorithms and Complexity*, vol. 277. Cham, Switzerland: Springer, 2012.
- [44] B. M. Waxman, "Routing of multipoint connections," *IEEE J. Sel. Areas Commun.*, vol. SAC-6, no. 9, pp. 1617–1622, Dec. 1988.
- [45] R. K. Jain, D.-M. W. Chiu, and W. R. Hawe, "A quantitative measure of fairness and discrimination," Eastern Res. Lab., Digit. Equip. Corp., Hudson, MA, USA, Tech. Rep. DEC-TR-301, 1984, vol. 21.
- [46] R. Van Meter et al., "A quantum Internet architecture," in *Proc. IEEE Int. Conf. Quantum Comput. Eng. (QCE)*, Jun. 2022, pp. 341–352.



Zhaoying Wang (Student Member, IEEE) received the bachelor's degree in information security from the School of Cyber Science and Technology, University of Science and Technology of China (USTC), in 2019, where he is currently pursuing the Ph.D. degree. His research interests include quantum internet architecture and quantum networking.



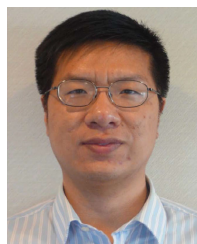
Jian Li (Senior Member, IEEE) received the bachelor's degree from the Department of Electronics and Information Engineering, Anhui University, in 2015, and the Ph.D. degree from the Department of Electronic Engineering and Information Science, University of Science and Technology of China (USTC), in 2020. From November 2019 to November 2020, he was a Visiting Scholar with the Department of Electronic and Computer Engineering, University of Florida. From December 2020 to December 2022, he was a Post-Doctoral Researcher with the School of Cyber Science and Technology, USTC. He is currently an Associate Researcher with the School of Cyber Science and Technology, USTC. His research interests include quantum networking and wireless networks. He serves as an Editor for *China Communications*.



Kaiping Xue (Senior Member, IEEE) received the bachelor's degree from the Department of Information Security, University of Science and Technology of China (USTC), in 2003, and the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), USTC, in 2007. From May 2012 to May 2013, he was a Post-Doctoral Researcher with the Department of Electrical and Computer Engineering, University of Florida. He is currently a Professor with the School of Cyber Science and Technology, USTC. He is also the Director of Network and Information Center, USTC. His research interests include next-generation internet architecture design, transmission optimization, and network security. He is an IET Fellow. His work won the Best Paper Awards from IEEE MSN 2017 and IEEE HotCN 2019, the Best Paper Honorable Mention from ACM CCS 2022, the Best Paper Runner-Up Award from IEEE MASS 2018, and the Best Track Paper from MSN 2020. He has served as the Track/Symposium Chair for some reputed conferences. He serves on the Editorial Board for several journals, including IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, IEEE TRANSACTIONS ON DEPENDABLE AND SECURE COMPUTING, IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, and IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT. He has served as a Guest Editor for many reputed journals/magazines, and he has also served as a track/symposium chair for some reputed conferences.



Zhonghui Li (Member, IEEE) received the bachelor's degree from the School of Information and Software Engineering, University of Electronic Science and Technology of China, in 2018, and the Ph.D. degree in cyber security from the School of Cyber Science and Technology, University of Science and Technology of China (USTC), in 2023. He is currently a Post-Doctoral Researcher with the School of Cyber Science and Technology, USTC. His current research interests include quantum networking and network security.



Ruidong Li (Senior Member, IEEE) received the bachelor's degree in engineering from Zhejiang University, China, in 2001, and the Doctor of Engineering degree from the University of Tsukuba in 2008. He is an Associate Professor with the College of Science and Engineering, Kanazawa University, Japan. Before joining Kanazawa University, he was a Senior Researcher with the Network System Research Institute, National Institute of Information and Communications Technology (NICT). His current research interests include future networks, big

data networking, blockchain, information-centric networks, the Internet of Things, network security, wireless networks, and quantum internet. He is a member of IEICE. He is the Founder and the Chair of the IEEE SIG on big data intelligent networking and the IEEE SIG on intelligent internet edge and the secretary of the IEEE Internet Technical Committee. He also serves as the Chair for conferences and workshops, and organizes the special issues for the leading magazines and journals.



Nenghai Yu received the bachelor's degree from Nanjing University of Posts and Telecommunications, Nanjing, China, in 1987, the M.E. degree from Tsinghua University, Beijing, China, in 1992, and the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), Hefei, China, in 2004. He is currently a Professor with the School of Cyber Science and Technology and the Department of Electronic Engineering and Information Science, USTC. He is the Executive Dean of the School of Cyber Science and Technology, USTC, and the Director of the Information Processing Center, USTC. He has authored or co-authored more than 130 papers in journals and international conferences. His research interests include multimedia security, multimedia information retrieval, video processing, and information hiding.



Qibin Sun (Fellow, IEEE) received the Ph.D. degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), in 1997. He is currently a Professor with the School of Cyber Science and Technology, USTC. He has published more than 120 papers in international journals and conferences. His research interests include multimedia security, network intelligence, and security.



Jun Lu received the bachelor's degree from Southeast University in 1985 and the master's degree from the Department of Electronic Engineering and Information Science (EEIS), University of Science and Technology of China (USTC), in 1988. He is currently a Professor with the School of Cyber Science and Technology and the Department of EEIS, USTC. He is also the President of Jiaxing University. His research interests include theoretical research and system development in the field of integrated electronic information systems, network, and information security. He is an Academician of Chinese Academy of Engineering (CAE).